

Pre-CRM AI-Powered Revenue Engine

DATA INGESTION RELIABILITY, STAGING, SIGNAL SCORING & SUB-5-MINUTE SPEED-TO-LEAD AUTOMATION

An enterprise-grade GTM engineering specification for capturing inbound leads via Hookdeck, staging prospects in Supabase (PostgreSQL), executing multi-stage data qualification, scoring ICP with Claude AI, and triggering automated SLA response flows before syncing to HubSpot CRM.

1. Executive Summary & System Architecture

In modern B2B RevOps, pushing raw, unverified lead forms directly into a commercial CRM creates severe pipeline pollution, inflates HubSpot contact tier costs, and degrades sales rep conversion velocity. The **Pre-CRM AI-Powered Revenue Engine** establishes an ingestion and staging layer inside **Hookdeck** and **Supabase (PostgreSQL)** to queue, clean, enrich, score, and gate prospects before any CRM record is created.

Core Module Breakdown

Module 0: Hookdeck Webhook Reliability Gateway

Acts as the entry point for all inbound webhooks (forms, site triggers, app events). Queues payload traffic, prevents n8n server overload during traffic spikes, handles automatic retries on failure, and throttles API execution.

Module 1: Ingestion & Sanitization (n8n)

Catches verified payloads from Hookdeck. JavaScript Code Nodes clean messy formatting, standardize domain syntax, and strip malformed text string inputs.

Module 2: Relational Staging (Supabase)

Writes raw data to `staged_leads` PostgreSQL tables prior to CRM contact creation. Holds unverified, cold, or low-scoring leads safely outside of paid HubSpot tiers.

Module 3: Claude AI Intelligence

n8n routes lead parameters to Claude API using structured XML system prompts. Outputs a 0–100 ICP score, buying intent rating, and custom personalized sales opening line.

Module 4: Smart Routing & Response

Evaluates ICP score: **Score ≥ 70** triggers HubSpot CRM deal creation, Slack rep alerts, and sub-5-minute AI outreach. **Score < 70** routes to Supabase automated nurture.

Strategic RevOps Impact: This architecture achieves 100% CRM pipeline hygiene, eliminates ghost leads, prevents webhook loss during outages, optimizes paid contact seat limits, and guarantees sales reps spend 100% of their time on verified, high-intent prospects.

2. Complete 7-Stage Pre-CRM Data Staging & Qualification Gate

Before a lead payload touches the AI scoring node or CRM sync engine, it passes sequentially through seven staging gates in Hookdeck, n8n, and Supabase:

0 Hookdeck Webhook Queue & Rate-Throttling Gate

Catches incoming raw webhooks from Webflow, Typeform, or App Events. Instantly acknowledges the submission (200 OK) to the frontend while queuing payloads to prevent downstream n8n or Claude API rate limits (429 errors).

Pass: Send payload to n8n at controlled rate limit

Downstream Outage: Hold in Hookdeck Dead-Letter Queue (DLQ) & retry automatically

1 Spam & Syntax Filter

Executes regex checks for valid email structures, blocks disposable email domain providers (e.g., `@tempmail.com`), and filters bot submissions.

Pass: Proceed to Stage 2

Fail: Drop & Log in Supabase Spam Table

2 Email Deliverability & Verification Gate

Triggers an automated API call to ZeroBounce / Hunter / NeverBounce to perform live SMTP pings. Ensures the mailbox exists and won't cause hard email bounces.

Deliverable: Proceed to Stage 3 **Invalid/Bounce:** Tag 'Unverified' in Supabase

3 Deduplication & Identity Stitching

Queries Supabase staged_leads and HubSpot CRM by email and primary domain key. If the contact exists, n8n appends the activity event (e.g., "Visited Pricing 3x") to the existing record instead of creating duplicates.

New Contact: Proceed to Stage 4 **Existing Contact:** Append Event Timeline

4 Waterfall Data Enrichment

To reduce form friction, frontend forms ask only for Work Email. n8n passes the domain through a waterfall query across Clay / Apollo / Clearbit APIs to automatically append company headcount, funding round, tech stack, and HQ location.

Enriched: Append Firmographics

5 Anti-ICP Hard Exclusion Filter

Applies hard disqualification logic: automatically rejects personal email domains (e.g., @gmail.com), unsupported jurisdictions, student submissions, or competitor IP blocks.

Qualified Fit: Proceed to AI Scoring **Hard Exclusion:** Mark 'Disqualified' in DB

6 AI ICP & Intent Scoring Engine (Claude LLM)

Passes enriched firmographic payloads, behavioral telemetry, and form comments to Claude API. Claude outputs a structured JSON response containing icp_score (0–100), buying_intent, and a personalized sales icebreaker.

Score ≥ 70: Promote to HubSpot CRM **Score < 70:** Supabase Nurture Flow

3. Sub-5-Minute "Speed-to-Lead" Response SLA Engine

Contacting an inbound lead within 5 minutes increases conversion rates by up to 8x compared to a 30-minute delay. The system integrates automated multi-channel response triggers conditioned on the lead's AI qualification score and contact preferences:

TARGET CHANNEL	TRIGGER CONDITIONS	AUTOMATION ENGINE	TARGET EXECUTION SLA
AI Outbound Voice Call	ICP Score ≥ 70 + Valid Phone + High Intent Signal	Vapi AI / Retell / Bland AI API via n8n	Under 30 Seconds
Interactive WhatsApp / SMS	ICP Score ≥ 70 + Mobile Preference / Secondary Channel	Twilio API / WhatsApp Business API	Under 45 Seconds
Context-Aware AI Email	All Qualified Leads (Score ≥ 70) or Email Preference	Claude API (Copy generation) + Resend API	Under 60 Seconds
Sales Rep Slack Escalation	High-Value Account (Target Deal > \$20k) / Custom Handoff	n8n Webhook → Slack #sales-hot-transfers	Instant Alert (< 15s)

Dynamic Overflow Triage with Human-in-the-Loop (HITL) Handoff

If assigned Account Executives (AEs) are currently engaged in client calls or marked busy in Supabase, the AI Sales Rep automatically activates to handle the prospect immediately. As soon as a human rep frees up, or if the prospect requests human assistance, n8n executes a seamless transfer:

- **AI State Paused:** AI agent status shifts to paused with full transcript preserved.
- **Contextual Rep Briefing:** Pushes a 3-bullet AI conversation summary, ICP score, and identified pain points to Slack so the rep never asks the buyer to repeat themselves.
- **One-Click Takeover:** The human rep clicks "Takeover Chat" in Retool or Slack to assume live control.

4. Technical JSON Payload Specifications

Below are the standard data packets transmitted across Hookdeck, the n8n pipeline, Supabase staging tables, and Claude AI LLM nodes:

1. Ingestion Payload with Hookdeck Metadata Layer

```
{
  "hookdeck_metadata": {
    "event_id": "evt_hk_9981a3b",
    "attempts": 1,
    "ingested_at": "2026-08-13T14:24:49Z",
    "source_id": "src_webflow_pricing_form"
  },
  "body": {
    "event": "lead.captured",
    "email": "alex@ai-labs.io",
    "company_name": "AI Labs Corp",
    "monthly_spend": "$20k+",
    "primary_provider": "anthropic",
    "timestamp": "2026-08-13T14:24:50Z"
  }
}
```

2. Claude AI Evaluation & Scoring Output

```
{
  "lead_id": "lead_9981",
  "icp_score": 88,
  "buying_intent": "High",
  "route_decision": "hubspot_crm",
  "personalized_icebreaker": "Notice AI Labs spends $20k+ on Anthropic. CostPulse can cap token spikes automatically."
}
```

3. Speed-to-Lead Execution Payload

```
{
  "event": "speed_to_lead.triggered",
  "lead_id": "lead_9981",
  "channel_activated": "vapi_voice_agent",
  "dial_status": "initiated",
  "latency_seconds": 24.2,
  "rep_overflow_status": "ae_busy_ai_active",
  "hubspot_deal_id": "deal_88301",
  "slack_alert_sent": true
}
```

5. Commercial RevOps Impact & Portfolio Positioning

Building and demonstrating this unified Pre-CRM AI Revenue Engine establishes immediate credibility with hiring managers, VPs of RevOps, and SaaS Founders:

Metric / Dimension	Standard Legacy Process	CostPulse AI Staged Engine (v2.0 with Hookdeck)	Net Business Impact
System Uptime & Ingestion	Webhook drops during server outages or traffic surges.	99.99% Guaranteed Ingestion via Hookdeck queueing & retries.	Zero Lead Leakage under traffic spikes.
Speed-to-Lead Response	2 to 24 Hours (Manual AE Assignment).	Under 30 Seconds (Automated Multi-Channel).	8x Higher Lead Conversion.
CRM Data Hygiene	30%–40% Duplicate / Bad Records.	100% Staged & Verified prior to sync.	Zero CRM Data Bloat.
HubSpot Tier Costs	Pays for all raw, unverified signups.	Only pays for qualified contacts (Score ≥ 70).	40%+ Cost Reduction.
Sales Rep Efficiency	50% time spent chasing unqualified leads.	100% focus on AI-scored high-intent leads.	Maximized AE Pipeline ARR.

Portfolio Value Proposition: Presenting this full architecture proves end-to-end engineering capability across ingestion gateways (Hookdeck), frontend web triggers (React/Vercel), automated workflow engines (n8n), staging databases (Supabase), AI intelligence APIs (Claude), commercial CRMs (HubSpot), and internal operational tools (Retool/Slack).